

CURRICULUM VITAE

Sishi Li (Dr)

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CURRENT POSITION

10/2025 – Now Research Fellow, Department of The Built Environment, National University of Singapore, Singapore

EDUCATION

08/2020 – 06/2025 Ph.D., Tsinghua University, Beijing, China

- Civil Engineering, Department of Building Science, GPA: 3.87/4.00
- Dissertation: Design and evaluation of a dynamic non-uniform thermal environment; Supervisor: Prof. Bin Cao

09/2016 – 06/2020 B.Eng., Chongqing University, Chongqing, China

- Building Environment and Energy Application Engineering, School of Civil Engineering, GPA: 3.84/4.00 (rank: 1/114)

RESEARCH INTERESTS

- Occupant-centric thermal control (e.g. hybrid cooling strategy, automation thermal control design)
- Dynamic & Personal thermal environment design (e.g., indoor dynamic temperature control, user-oriented thermal environment conditioning)
- Thermal protection in extremely cold environments (e.g., thermal protection against extreme cold exposure and large temperature drops)

SKILLS

Software ▪ Data analysis: R, MATLAB, Python, SPSS ▪ Drawing/Modeling: AutoCAD ▪ Others: Microsoft Office, DeST, Airpak

Experiments ▪ Field tests: Field environment measurement, HVAC system performance tests, Field survey design

▪ Climate chamber experiments: Subjective experiments & Thermal manikin tests

HONORS & AWARDS

2024 Second prize of China Thermal Comfort Conference for Student Forum

2023 First-class Scholarship of Tsinghua University

2021 & 2024 Second-class Scholarship of Tsinghua University

2020 Excellent Graduate of Chongqing University

2019 National Encouragement Scholarship of China for Undergraduate Students

2017 National Scholarship of China for Undergraduate Students

2017 – 2019 First-class Excellent Student Scholarship of Chongqing University

RESEARCH PROJECT EXPERIENCES

National Key Research and Development Program of China (2018YFF0300200) 11/2022 – Now

- Research: Personal environmental comfort demand analysis and system design in non-uniform space-time environments (PI: Bin Cao)
- Role: Leading Researcher and Project Management of Sub-project

Tsinghua University-Toyota Joint Research Fund 11/2022 – 11/2024

- Research: Humanized design and products for the user interface in the built environment (PI: Bin Cao)
 - Role: Leading Researcher and Project Management of Sub-project
- National Key Research and Development Program of China (2022YFC3801500) 08/2018 – 12/2021
- Research: Thermal protection technology for winter outdoor spectating environment (PI: Bin Cao)
 - Role: Primary Researcher

PUBLICATIONS

Selected Published Papers:

- [1] S. Li, X. Jia, C. Peng, et al. Thermal comfort benefits, energy efficiency, and occupant regulation behaviour in four types of personal heating within the built environment, *Building and Environment* 266 (2024) 112075.
- [2] S. Li, X. Jia, C. Peng, et al. Effects of temperature cycles on human thermal comfort in built environment under summer conditions, *Science of The Total Environment* 912 (2024) 168756.
- [3] S. Li, X. Jia, B. Cao, et al. Improvement of thermal perceptions and physiological responses using torso heating under short-term cold exposure, *Science of The Total Environment* 892 (2023) 164491.
- [4] S. Li, X. Jia, B. Cao, et al. Thermal comfort characteristics and heating demand of people with different activity status during extremely cold exposure, *Building and Environment* 228 (2023) 109798.
- [5] S. Li, Y. Deng, B. Cao, Study on the Performance of Personal Heating in Extremely Cold Environments Using a Thermal Manikin, *Buildings* 13 (2023) 362.
- [6] X. Jia, S. Li, Y. Zhu, et al. Development of the Universal Standard Effective Temperature for evaluating thermal comfort across different metabolic rates, *Building and Environment* 250 (2024) 111149.
- [7] X. Jia, S. Li, J. Wang, et al. Measurement of steady state and dynamic metabolic rates under typical daily activities, *Building and Environment* 254 (2024) 111344.
- [8] X. Jia, S. Li, Y. Zhu, et al. Transient thermal comfort and physiological responses following a step change in activity status under summer indoor environments, *Energy and Buildings* 285 (2023) 112918.
- [9] S. Li, B. Cao, Study on Wearable Personalized Thermal Comfort System for Cold Environments, *World Architecture* 6 (2022) 61-65. (in Chinese)
- [10] S. Li, M. Zhao, Y. Zhai, et al. Optimization Study of Deflecting Ventilation Based on CFD and Orthogonal Tests, *Building Energy & Environment* 8 (2020) 25-30. (in Chinese)

CONFERENCES

- [1] S. Li, B. Cao, Y. Zhu. The effects of temperature cycles on thermal comfort in built environment under summer conditions. *Thermal Comfort Conference and the 5th Academic Forum of the Thermal Comfort Group and the Sleep Environment and Health Group in China*, November 15-17, 2024.
- [2] S. Li, B. Cao, Y. Zhu. Thermal performance evaluation of four personal heating devices and occupant usage behaviour under three cold conditions. *Indoor Air 2024*, July 7-11, 2024, Honolulu, USA.
- [3] S. Li, B. Cao, Y. Zhu. Human responses to temperature cycles under summer conditions. *The 13th International Symposium on Heating, Ventilation, and Air Conditioning (ISHVAC 2023)*, November 14-16, 2023, Beijing, China.
- [4] S. Li, B. Cao. Thermal perception and physiological response of people with different heating modes during short-term cold exposure. *Healthy Buildings 2023 Asia and Pacific Rim*, July 17-19, 2023, Tianjin, China.
- [5] S. Li, B. Cao, Y. Zhu. Effect of activity status on thermal comfort during extremely cold exposure. *Indoor Air 2022*, June 12-16, 2022, Kuopio, Finland.